

GENIFY

LECTURE 04

23-July-2025

TOPICS COVERED:

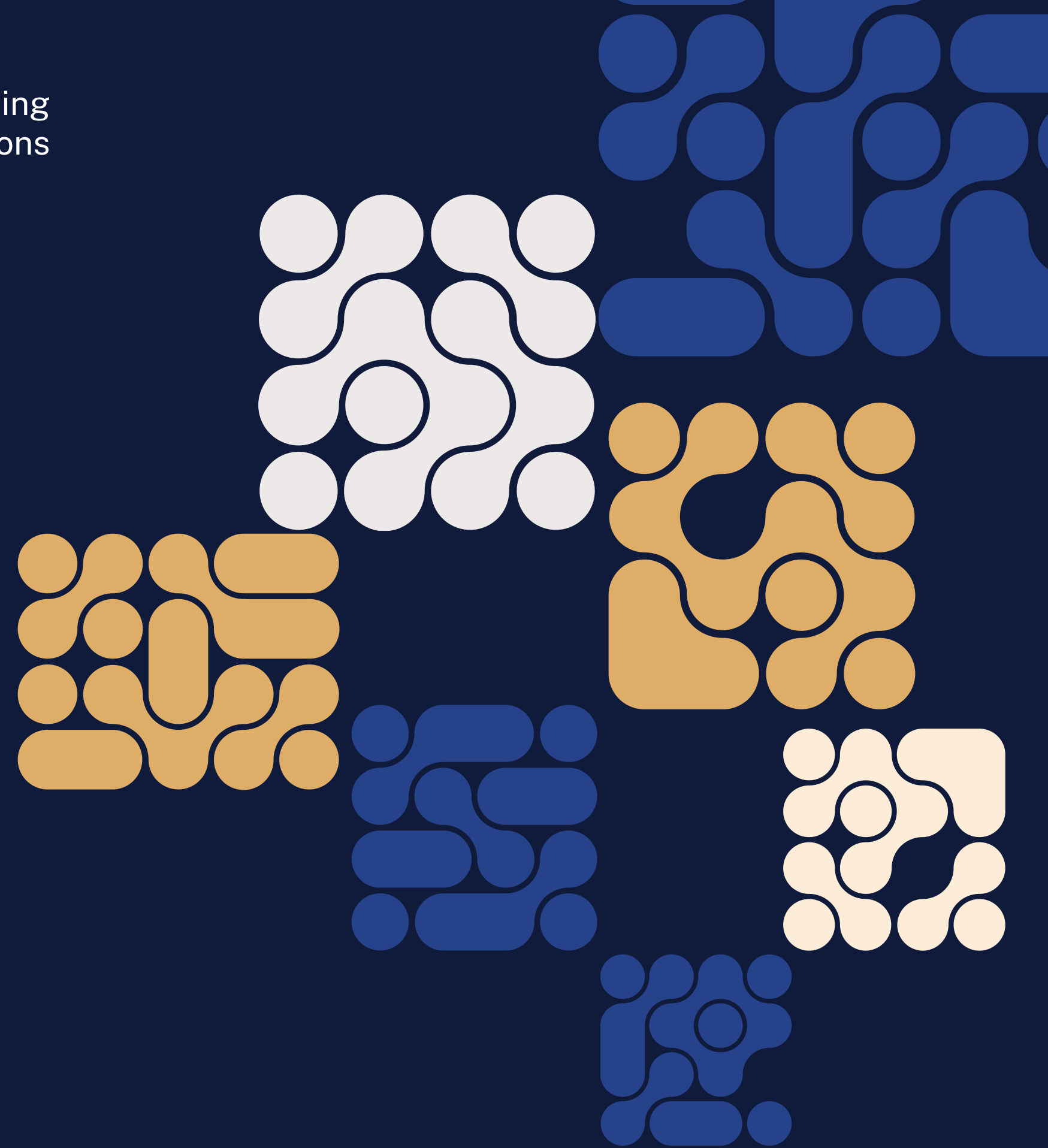
- MACHINE LEARNING
 - TYPES OF ML
 - SUPERVISED ML
 - UNSUPERVISED ML
 - REINFORCEMENT ML

Presenter

Shahzaib Kashif

Co-Presenter

Shayan Hassan Baig



RECAP QUIZ TIME

Home Tasks

Discussion

T1: Convert Age and Embark columns into numerical values in Titanic Dataset

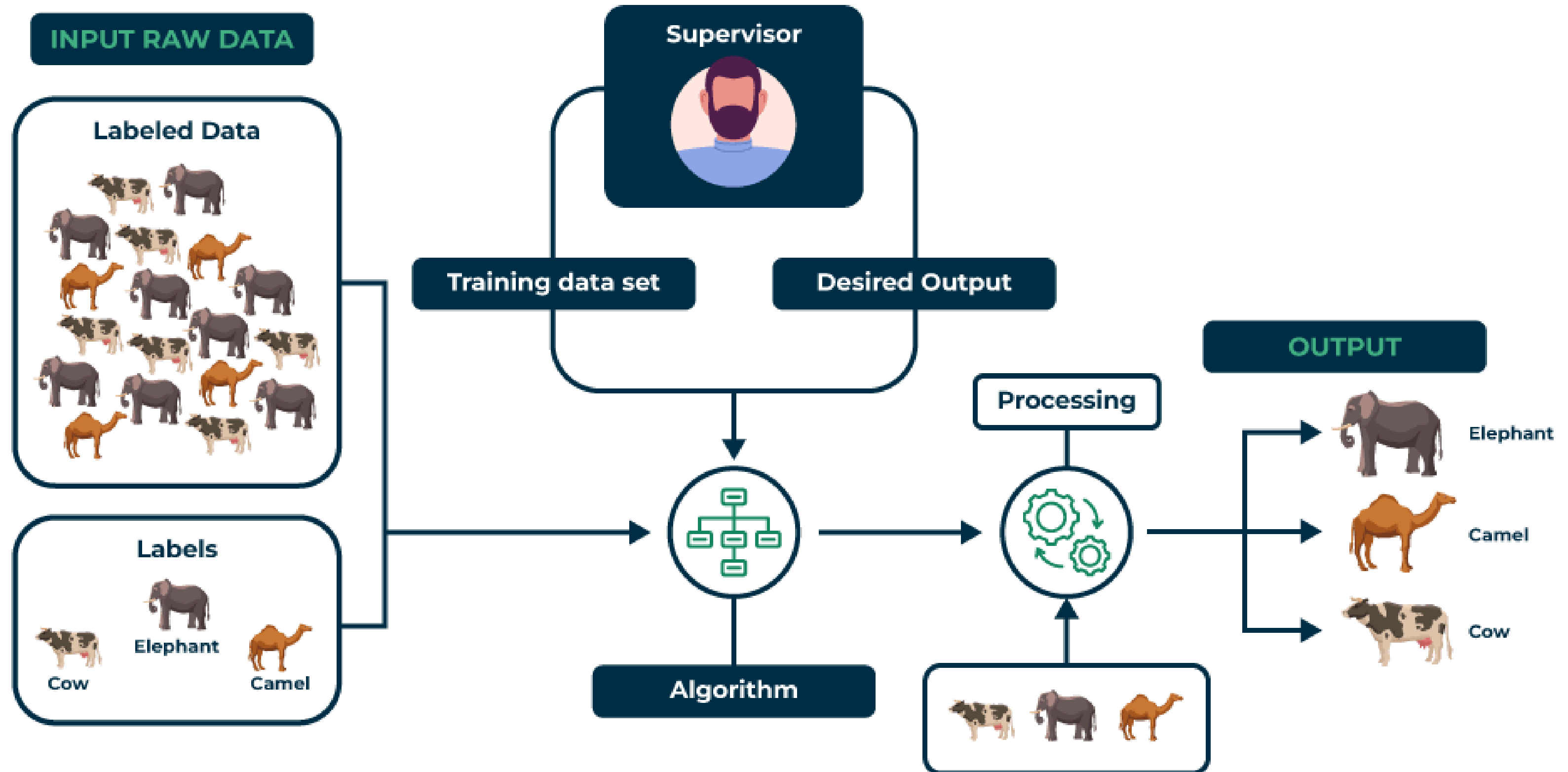
T2: Which Feature Engineering should apply to the Diabetes Dataset (MinMaxScaler OR Z-Score Standarization)



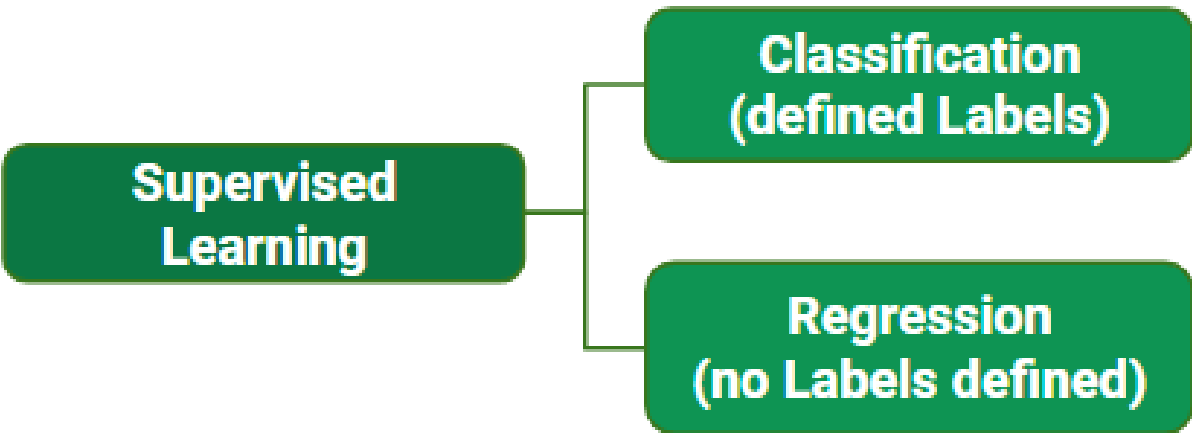
MACHINE LEARNING

Supervised Learning

Supervised Learning



Types of Supervised ML



- Supervised learning algorithms are generally categorized into two main types:
 - Classification - where the goal is to predict discrete labels or categories
 - Regression - where the aim is to predict continuous numerical values.
- While training the model, data is usually split the ration to 80:20 i.e. 80% as training data and the rest as testing data.
- In training data, we feed input as well as output for 80% of data.
- The model learns from training data only. We use different supervised learning algorithms (which we will discuss in detail in the next section) to build our model.
- Let's first understand the classification and regression data through the table below:

Figure A: It is a dataset of a shopping store that is useful in predicting whether a customer will purchase a particular product under consideration or not based on his/ her gender, age, and salary.

Input: Gender, Age, Salary

Output: Purchased i.e. 0 or 1; 1 means yes the customer will purchase and 0 means that the customer won't purchase it.

Figure B: It is a Meteorological dataset that serves the purpose of predicting wind speed based on different parameters.

Input: Dew Point, Temperature, Pressure, Relative Humidity, Wind Direction

Output: Wind Speed

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

Practical Examples of Supervised ML

- **Fraud Detection in Banking:** Utilizes supervised learning algorithms on historical transaction data, training models with labeled datasets of legitimate and fraudulent transactions to accurately predict fraud patterns
- **Parkinson Disease Prediction:** Parkinson's disease is a progressive disorder that affects the nervous system and the parts of the body controlled by the nerves.
- **Customer Churn Prediction:** Uses supervised learning techniques to analyze historical customer data, identifying features associated with churn rates to predict customer retention effectively.
- **Cancer cells classification:** Implements supervised learning for cancer cells based on their features, and identifying them if they are 'malignant' or 'benign'.
- **Stock price prediction:** Applies supervised learning to predict a signal that indicates whether buying a particular stock will be helpful or not.

Common Supervised ML Algortihms

Algorithm	Regression, Classification	Purpose	Method	Use Cases
Linear Regression	Regression	Predict continuous output values	Linear equation minimizing sum of squares of residuals	Predicting continuous values
Logistic Regression	Classification	Predict binary output variable	Logistic function transforming linear relationship	Binary classification tasks
Decision Trees	Both	Model decisions and outcomes	Tree-like structure with decisions and outcomes	Classification and Regression tasks
Random Forests	Both	Improve classification and regression accuracy	Combining multiple decision trees	Reducing overfitting, improving prediction accuracy
SVM	Both	Create hyperplane for classification or predict continuous values	Maximizing margin between classes or predicting continuous values	Classification and Regression tasks
KNN	Both	Predict class or value based on k closest neighbors	Finding k closest neighbors and predicting based on majority or average	Classification and Regression tasks, sensitive to noisy data
Gradient Boosting	Both	Combine weak learners to create strong model	Iteratively correcting errors with new models	Classification and Regression tasks to improve prediction accuracy
Naive Bayes	Classification	Predict class based on feature independence assumption	Bayes' theorem with feature independence assumption	Text classification, spam filtering, sentiment analysis, medical

Training a Supervised Learning Model: Key Steps

The goal of Supervised learning is to generalize well to unseen data. Training a model for supervised learning involves several crucial steps, each designed to prepare the model to make accurate predictions or decisions based on labeled data. Below are the key steps involved in training a model for supervised machine learning:

- **Data Collection and Preprocessing:** Gather a labeled dataset consisting of input features and target output labels. Clean the data, handle missing values, and scale features as needed to ensure high quality for supervised learning algorithms.
- **Splitting Data:** Divide the data into training set (80%) and test set (20%).
- **Choosing the Model:** Select appropriate algorithms based on the problem type. This step is crucial for effective supervised learning in AI.
- **Training the Model:** Feed the model input data and output labels, allowing it to learn patterns by adjusting internal parameters.
- **Evaluating the model:** Test the trained model on the unseen test set and assess its performance using various metrics.
- **Hyperparameter Tuning:** Adjust settings that control the training process (e.g., learning rate) using techniques like grid search and cross-validation.
- **Final Model Selection and Testing:** Retrain the model on the complete dataset using the best hyperparameters testing its performance on the test set to ensure readiness for deployment.
- **Model Deployment:** Deploy the validated model to make predictions on new, unseen data.

Linear Regression

- Linear regression is a type of supervised machine learning that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets.
- It assumes that there is a linear relationship between the input and output, meaning the output changes at a constant rate as the input changes.
- This relationship is represented by a straight line.
- **For example:** we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied.
Here
 - **Independent variable (input):** Hours studied because it's the factor we control or observe.
 - **Dependent variable (output):** Exam score because it depends on how many hours were studied.
- We use the independent variable to predict the dependent variable.

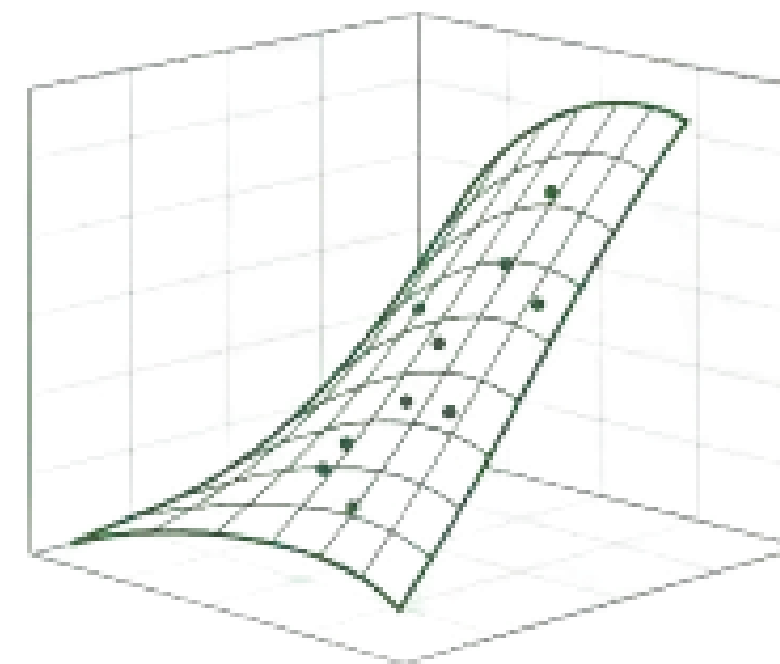
Types of Linear Regression

Simple Linear Regression



Predicts the dependent variable using a single independent variable.

Multiple Linear Regression



Uses two or more independent variables to predict the dependent variable.

Real-World Use Cases of Linear Regression

01

Stock
Market
Prediction



02

Real Estate
Price
Prediction



03

Medical Risk
Prediction



04

Sales
Forecasting



Python Walkthrough

<https://github.com/merledu/genify/blob/main/lecture4/SimpleLinearRegression.ipynb>



Types of Linear Regressions

1. Simple Linear Regression

A straight-line model that tries to predict a target using one input (feature).

Predict a person's weight using just their height: $Weight = a \cdot Height + b$ (Just one X $\rightarrow$ one Y)

2. Multiple Linear Regression

Like simple linear regression, but now using two or more input features.

Predict a house's price using its size, number of rooms, and location.

$Price = a_1 \cdot Size + a_2 \cdot Rooms + a_3 \cdot Location + b$ (Multiple Xs $\rightarrow$ one Y.)

3. Polynomial Regression

Still linear in terms of learning, but adds powers of the input to capture curves, not just straight lines.

Predict car speed vs. time, where the relation forms a curve.

$Speed = a_1 \cdot t + a_2 \cdot t^2 + a_3 \cdot t^3 + b$ [useful when the data is non-linear but still smooth]

4. Ridge Regression

Like multiple regression, but it adds a penalty for large coefficients to reduce overfitting.

If your model is using too many features and overfitting, ridge helps shrink their effect to make the model generalize better.

5. Lasso Regression

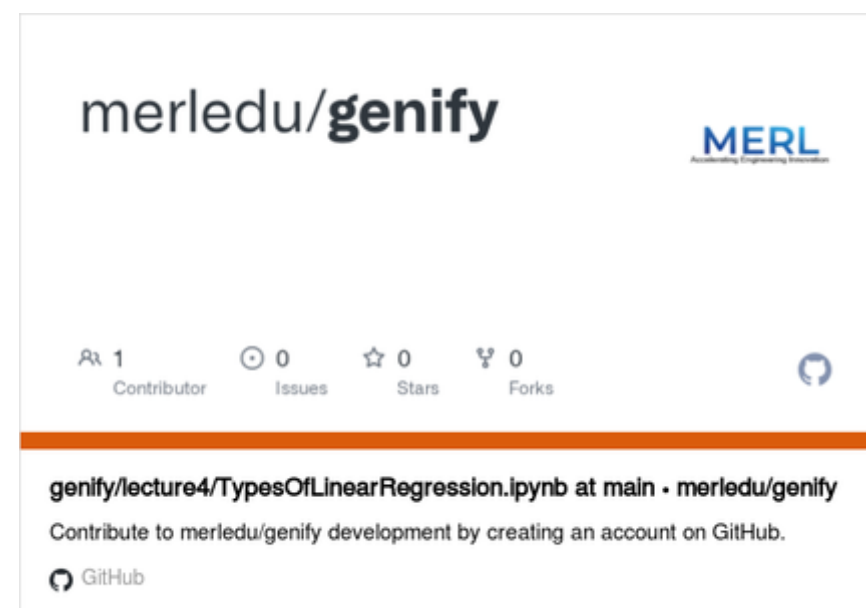
Like ridge, but uses absolute values instead of squares in the penalty.

It can completely remove some features by setting their weights to zero.

Good for when you want to select the most important features.

Python Walkthrough

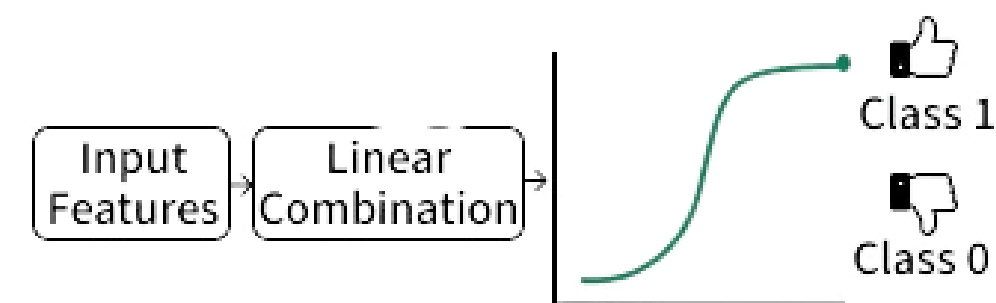
<https://github.com/merledu/genify/blob/main/lecture4/TypesOfLinearRegression.ipynb>



Logistic Regression

- Logistic Regression is a supervised machine learning algorithm used for classification problems.
- Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class.
- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

- Predicts the probability of a binary outcome (Yes/No, 0/1)
- Uses the sigmoid function to map inputs to probabilities (0 to 1)
- Ideal for classification tasks



Assumptions of Logistic Regression

1. **Independent observations:** Each data point is assumed to be independent of the others means there should be no correlation or dependence between the input samples.
2. **Binary dependent variables:** It takes the assumption that the dependent variable must be binary, means it can take only two values. For more than two categories SoftMax functions are used.
3. **Linearity relationship between independent variables and log odds:** The model assumes a linear relationship between the independent variables and the log odds of the dependent variable which means the predictors affect the log odds in a linear way.
4. **No outliers:** The dataset should not contain extreme outliers as they can distort the estimation of the logistic regression coefficients.
5. **Large sample size:** It requires a sufficiently large sample size to produce reliable and stable results.

Understanding Sigmoid Function

1. The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.
2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.
3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.
 - If the sigmoid output is same or above the threshold, the input is classified as Class 1.
 - If it is below the threshold, the input is classified as Class 0.

This approach helps to transform continuous input values into meaningful class predictions.

Types of Logistic Regressions

1. Binary Logistic Regression

- Target variable has 2 classes
- E.g., Spam or Not Spam, Sick or Healthy

2. Multinomial Logistic Regression

- Target variable has 3+ classes, no natural order
- E.g., Red, Blue, Green

3. Ordinal Logistic Regression

- Target variable has ordered categories
- E.g., Low, Medium, High

Python Walkthrough

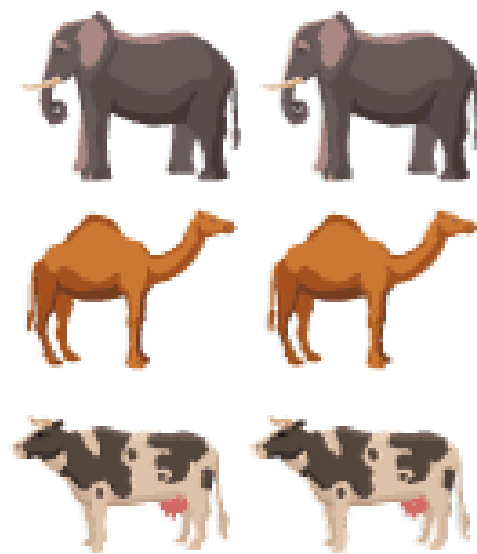
<https://github.com/merledu/genify/blob/main/lecture4/LogisticRegression.ipynb>



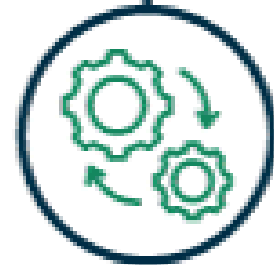
Unsupervised Learning

Unsupervised Learning

INPUT RAW DATA

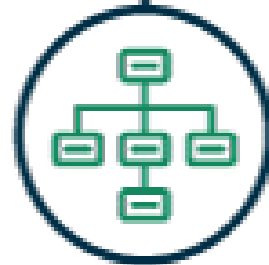


Interpretation

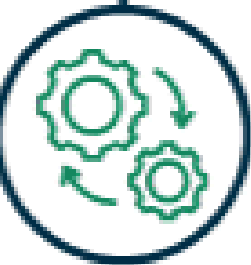


- Unknown Output
- No Training Data Set

Algorithm

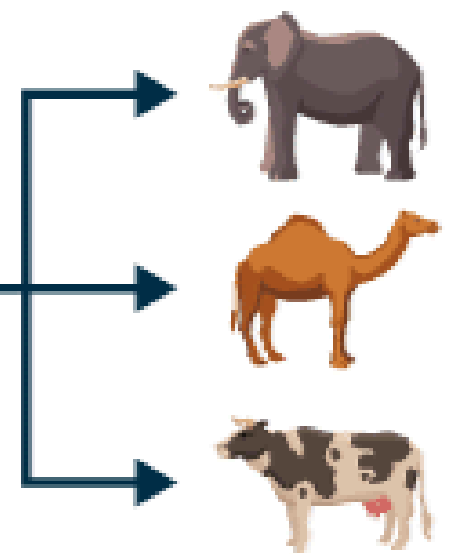


Processing



MODEL TRAINING

OUTPUT



What is Unsupervised ML?

- Unsupervised learning is a branch of machine learning that deals with unlabeled data.
- Unlike supervised learning, where the data is labeled with a specific category or outcome, unsupervised learning algorithms are tasked with finding patterns and relationships within the data without any prior knowledge of the data's meaning.
- Unsupervised machine learning algorithms find hidden patterns and data without any human intervention, i.e., we don't give output to our model.
- The training model has only input parameter values and discovers the groups or patterns on its own.

How does unsupervised learning work?

- Unsupervised learning works by analyzing unlabeled data to identify patterns and relationships.
- The data is not labeled with any predefined categories or outcomes, so the algorithm must find these patterns and relationships on its own.
- This can be a challenging task, but it can also be very rewarding, as it can reveal insights into the data that would not be apparent from a labeled dataset.
- Data-set in Figure A is Mall data that contains information about its clients that subscribe to them.
- Once subscribed they are provided a membership card and the mall has complete information about the customer and his/her every purchase.
- Now using this data and unsupervised learning techniques, the mall can easily group clients based on the parameters we are feeding in.

The input to the unsupervised learning models is as follows:

- **Unstructured data:** May contain noisy(meaningless) data, missing values, or unknown data
- **Unlabeled data:** Data only contains a value for input parameters, there is no targeted value(output). It is easy to collect as compared to the labeled one in the Supervised approach.

CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40
6	Female	22	17	76
7	Female	35	18	6
8	Female	23	18	94
9	Male	64	19	3
10	Female	30	19	72
11	Male	67	19	14
12	Female	35	19	99
13	Female	58	20	15
14	Female	24	20	77
15	Male	37	20	13
16	Male	22	20	79
17	Female	35	21	35

Figure A

Types of Unsupervised ML

- There are mainly 3 types of Algorithms which are used for Unsupervised dataset.
 - Clustering
 - Association Rule Learning
 - Dimensionality Reduction

Example: Imagine you have customer data (age, income, spending habits). Clustering can group customers into:

- Budget Shoppers
- Mid-range Shoppers
- Premium Shoppers

1. Clustering Algorithms

- Clustering in unsupervised machine learning is the process of grouping unlabeled data into clusters based on their similarities.
- The goal of clustering is to identify patterns and relationships in the data without any prior knowledge of the data's meaning.
- Broadly this technique is applied to group data based on different patterns, such as similarities or differences, our machine model finds.
- These algorithms are used to process raw, unclassified data objects into groups.
- For example, in the previous figure, we have not given output parameter values, so this technique will be used to group clients based on the input parameters provided by our data.

Some common Clustering Algorithms

- K-means Clustering: Groups data into K clusters based on how close the points are to each other.
- Hierarchical Clustering: Creates clusters by building a tree step-by-step, either merging or splitting groups.
- Density based Clustering (DBSCAN): Finds clusters in dense areas and treats scattered points as noise.
- Mean Shift Clustering: Discovers clusters by moving points toward the most crowded areas.
- Spectral Clustering: Groups data by analyzing connections between points using graphs.

Use: Market basket analysis

“If a customer buys bread and butter, they are likely to buy jam too.”

Example Rules:

{Milk, Cereal} → {Bananas}

(If someone buys milk and cereal, they are likely to buy bananas too.)

2. Association Rule Learning

- Association Rule learning is also known as association rule mining is a common technique used to discover associations in unsupervised machine learning.
- This technique is a rule-based ML technique that finds out some very useful relations between parameters of a large data set.
- This technique is basically used for market basket analysis that helps to better understand the relationship between different products.
- For e.g. shopping stores use algorithms based on this technique to find out the relationship between the sale of one product w.r.t to another's sales based on customer behavior. Like if a customer buys milk, then he may also buy bread, eggs, or butter. Once trained well, such models can be used to increase their sales by planning different offers.

Some common Association Rule Learning Algorithms

- Apriori Algorithm: Finds patterns by exploring frequent item combinations step-by-step.
- FP-Growth Algorithm: An Efficient Alternative to Apriori. It quickly identifies frequent patterns without generating candidate sets.
- Eclat Algorithm: Uses intersections of itemsets to efficiently find frequent patterns.
- Efficient Tree-based Algorithms: Scales to handle large datasets by organizing data in tree structures.

What? Reducing the number of input features (columns) in your dataset while keeping as much important information as possible.

Goal: Simplify data, Remove noise, Speed up training,

Example: Imagine a dataset with 100 features per customer. PCA might reduce this to 2 or 3 features that explain most of the variance, allowing you to plot and understand it more easily.

3. Dimensionality Reduction

- Dimensionality reduction is the process of reducing the number of features in a dataset while preserving as much information as possible.
- This technique is useful for improving the performance of machine learning algorithms and for data visualization.
- Imagine a dataset of 100 features about students (height, weight, grades, etc.).
- To focus on key traits, you reduce it to just 2 features: height and grades, making it easier to visualize or analyze the data.

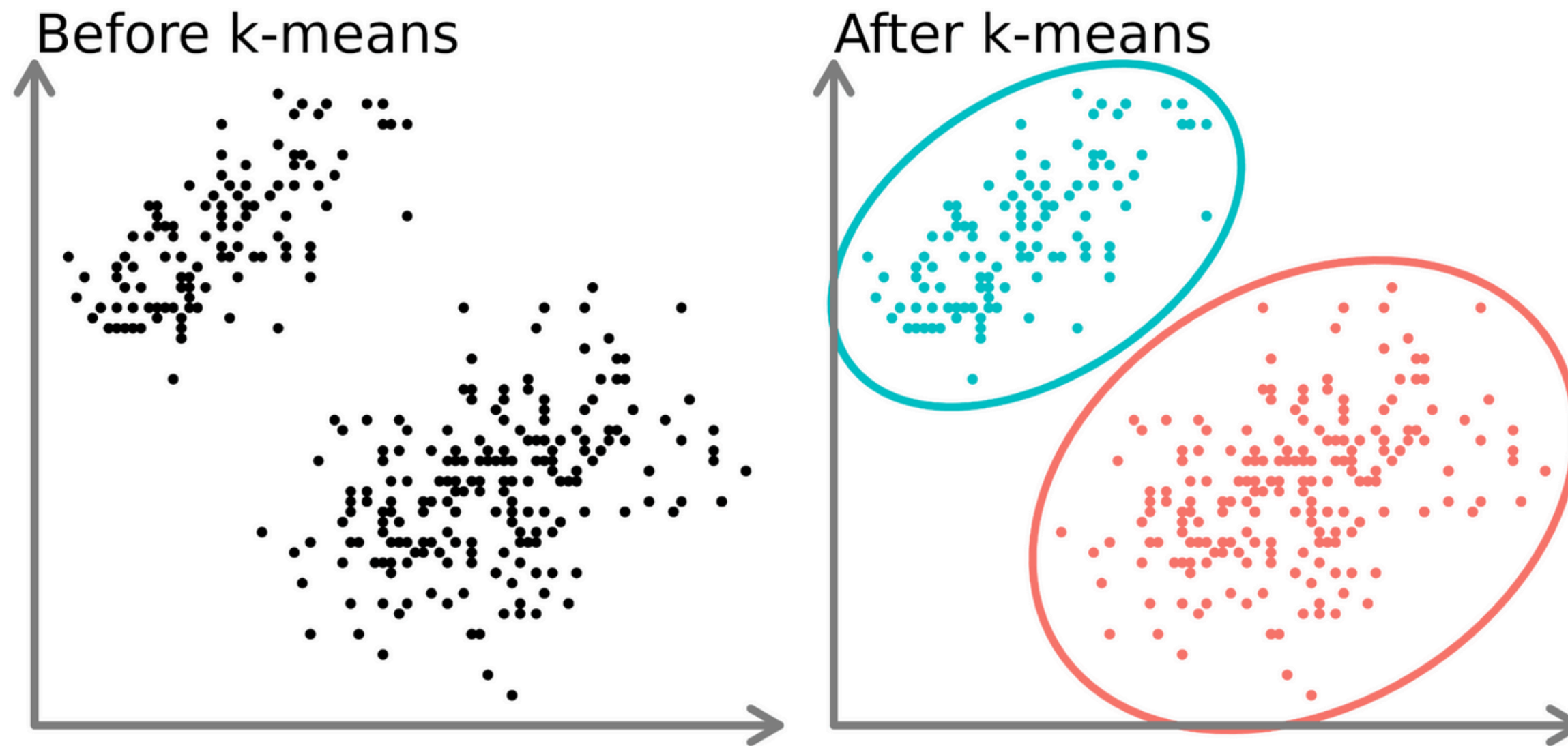
Some common Dimensionality Reduction Algorithms

- Principal Component Analysis (PCA): Reduces dimensions by transforming data into uncorrelated principal components.
- Linear Discriminant Analysis (LDA): Reduces dimensions while maximizing class separability for classification tasks.
- Non-Negative Matrix Factorization (NMF): Breaks data into non-negative parts to simplify representation.
- Locally Linear Embeddings (LLE): Reduces dimensions while preserving the relationships between nearby points.
- Isomap: Captures global data structure by preserving distances along a manifold.

Applications of Unsupervised learning

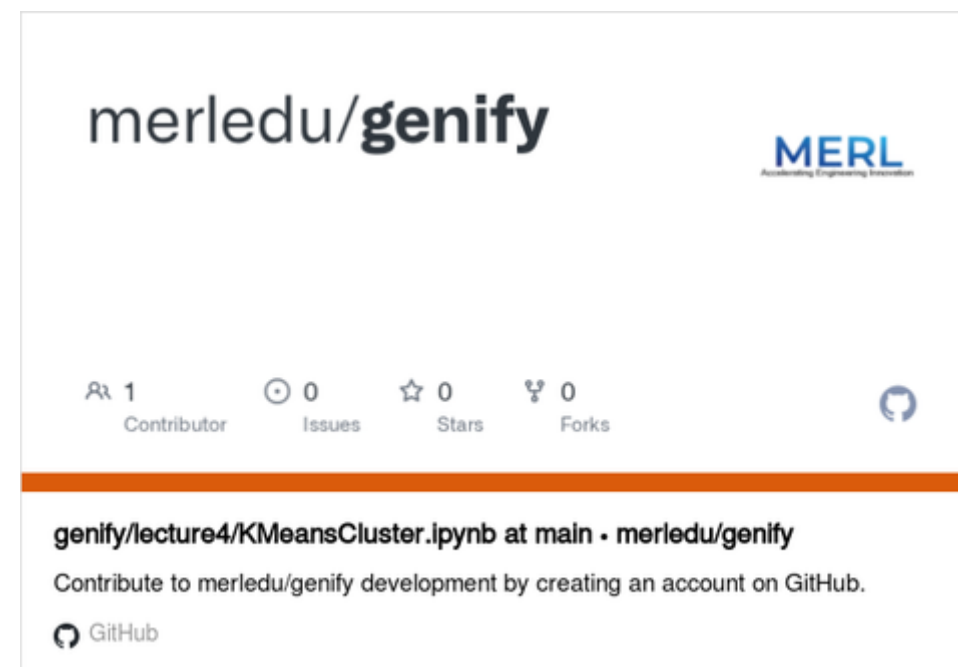
- *Customer Segmentation:* Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.
- *Anomaly Detection:* Identifies unusual patterns in data, aiding fraud detection, cybersecurity, and equipment failure prevention.
- *Recommendation Systems:* Suggests products, movies, or music by analyzing user behavior and preferences.
- *Image and Text Clustering:* Groups similar images or documents for tasks like organization, classification, or content recommendation.
- *Social Network Analysis:* Detects communities or trends in user interactions on social media platforms.
- *Astronomy and Climate Science:* Classifies galaxies or groups weather patterns to support scientific research

K Means Clustering

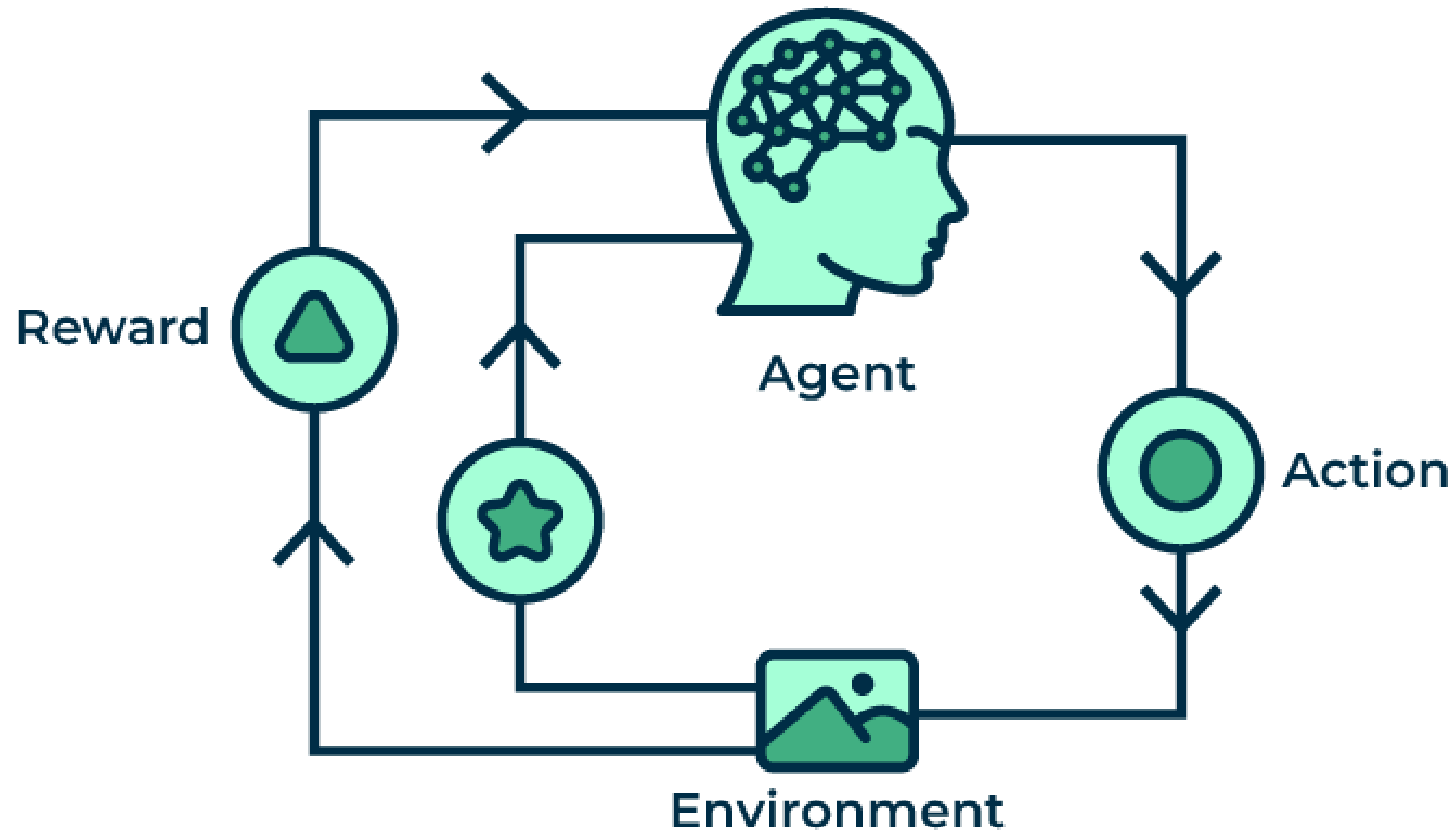


Python Walkthrough

<https://github.com/merledu/genify/blob/main/lecture4/KMeansCluster.ipynb>



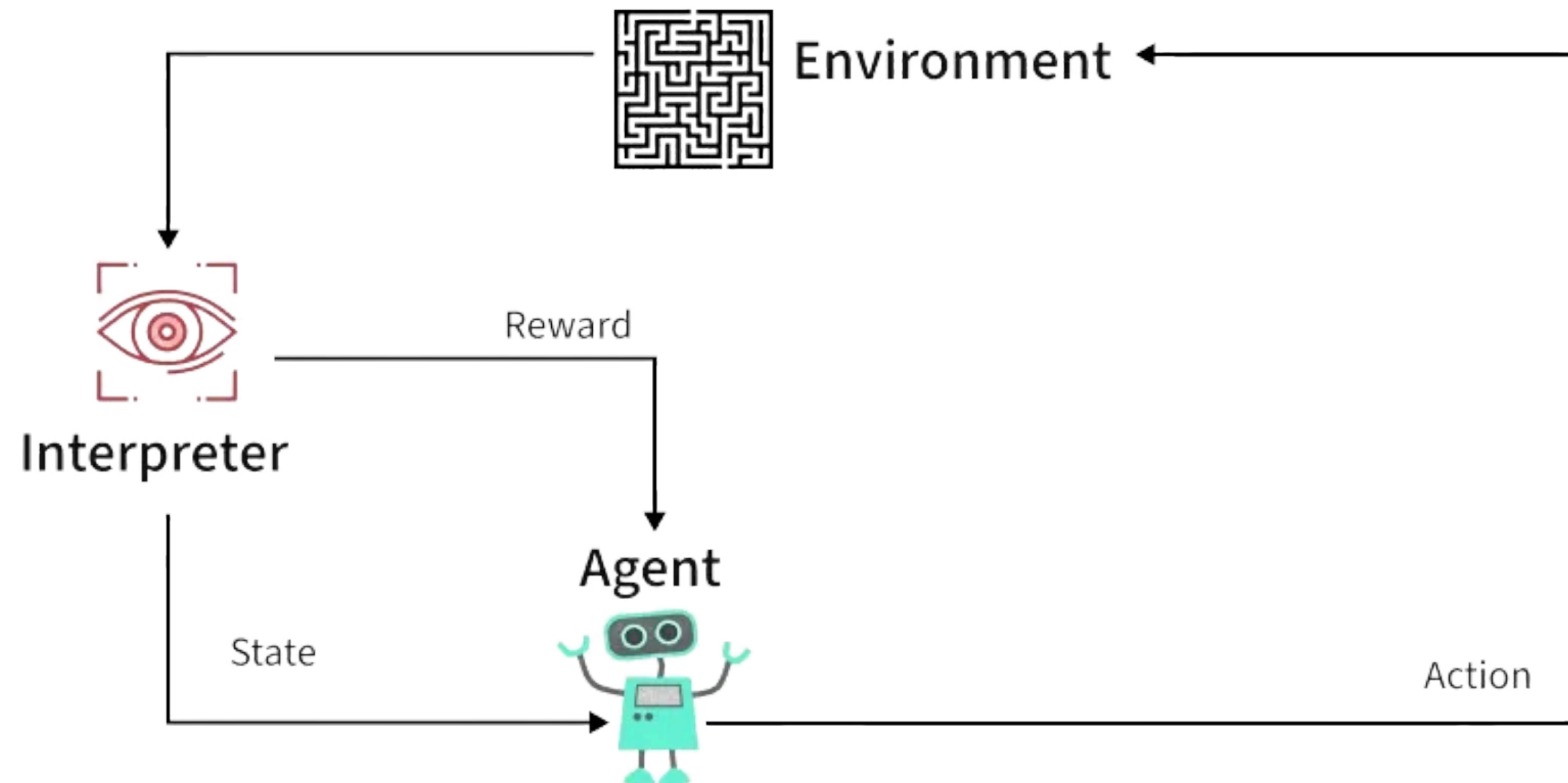
Reinforcement Learning



What is Reinforcement ML?

- Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards.
- RL allows machines to learn by interacting with an environment and receiving feedback based on their actions.
- This feedback comes in the form of rewards or penalties.

- Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal.
- The agent performs actions and receives feedback to optimize its decision-making over time.
 - Agent: The decision-maker that performs actions.
 - Environment: The world or system in which the agent operates.
 - State: The situation or condition the agent is currently in.
 - Action: The possible moves or decisions the agent can make.
 - Reward: The feedback or result from the environment based on the agent's action.



How Reinforcement Learning Works?

- The RL process involves an agent performing actions in an environment, receiving rewards or penalties based on those actions, and adjusting its behavior accordingly.
- This loop helps the agent improve its decision-making over time to maximize the cumulative reward.
- Here's a breakdown of RL components:
 - **Policy:** A strategy that the agent uses to determine the next action based on the current state.
 - **Reward Function:** A function that provides feedback on the actions taken, guiding the agent towards its goal.
 - **Value Function:** Estimates the future cumulative rewards the agent will receive from a given state.
 - **Model of the Environment:** A representation of the environment that predicts future states and rewards, aiding in planning.

Types of Reinforcement learning

1. Positive Reinforcement

- Positive Reinforcement is defined as when an event, occurs due to a particular behavior, increases the strength and the frequency of the behavior. In other words, it has a positive effect on behavior.
- **Advantages:** Maximizes performance, helps sustain change over time.
- **Disadvantages:** Overuse can lead to excess states that may reduce effectiveness.

2. Negative Reinforcement

- Negative Reinforcement is defined as strengthening of behavior because a negative condition is stopped or avoided.
- **Advantages:** Increases behavior frequency, ensures a minimum performance standard.
- **Disadvantages:** It may only encourage just enough action to avoid penalties.

Reinforcement Learning Example: Navigating a Maze

- Imagine a robot navigating a maze to reach a diamond while avoiding fire hazards. The goal is to find the optimal path with the least number of hazards while maximizing the reward:
- Each time the robot moves correctly, it receives a reward.
- If the robot takes the wrong path, it loses points.
- The robot learns by exploring different paths in the maze. By trying various moves, it evaluates the rewards and penalties for each path. Over time, the robot determines the best route by selecting the actions that lead to the highest cumulative reward.

The robot's learning process can be summarized as follows:

- **Exploration:** The robot starts by exploring all possible paths in the maze, taking different actions at each step (e.g., move left, right, up, or down).
- **Feedback:** After each move, the robot receives feedback from the environment:
 - A positive reward for moving closer to the diamond.
 - A penalty for moving into a fire hazard.
- **Adjusting Behavior:** Based on this feedback, the robot adjusts its behavior to maximize the cumulative reward, favoring paths that avoid hazards and bring it closer to the diamond.
- **Optimal Path:** Eventually, the robot discovers the optimal path with the least number of hazards and the highest reward by selecting the right actions based on past experiences.



Cart Pole in OpenAI Gym

- One of the classic RL problems is the **CartPole environment in OpenAI Gym**, where the goal is to balance a pole on a cart.
- The agent can either push the cart left or right to prevent the pole from falling over.
- **State space**: Describes the four key variables (position, velocity, angle, angular velocity) of the cart-pole system.
- **Action space**: Discrete actions—either move the cart left or right.
- **Reward**: The agent earns 1 point for each step the pole remains balanced.

Python Walkthrough

Application of Reinforcement Learning

- **Robotics:** RL is used to automate tasks in structured environments such as manufacturing, where robots learn to optimize movements and improve efficiency.
- **Game Playing:** Advanced RL algorithms have been used to develop strategies for complex games like chess, Go, and video games, outperforming human players in many instances.
- **Industrial Control:** RL helps in real-time adjustments and optimization of industrial operations, such as refining processes in the oil and gas industry.
- **Personalized Training Systems:** RL enables the customization of instructional content based on an individual's learning patterns, improving engagement and effectiveness.

Advantages of Reinforcement Learning

- **Solving Complex Problems:** RL is capable of solving highly complex problems that cannot be addressed by conventional techniques.
- **Error Correction:** The model continuously learns from its environment and can correct errors that occur during the training process.
- **Direct Interaction with the Environment:** RL agents learn from real-time interactions with their environment, allowing adaptive learning.
- **Handling Non-Deterministic Environments:** RL is effective in environments where outcomes are uncertain or change over time, making it highly useful for real-world applications.

Disadvantages of Reinforcement Learning

- **Not Suitable for Simple Problems:** RL is often an overkill for straightforward tasks where simpler algorithms would be more efficient.
- **High Computational Requirements:** Training RL models requires a significant amount of data and computational power, making it resource-intensive.
- **Dependency on Reward Function:** The effectiveness of RL depends heavily on the design of the reward function. Poorly designed rewards can lead to suboptimal or undesired behaviors.
- **Difficulty in Debugging and Interpretation:** Understanding why an RL agent makes certain decisions can be challenging, making debugging and troubleshooting complex.

1. Model-Based Methods

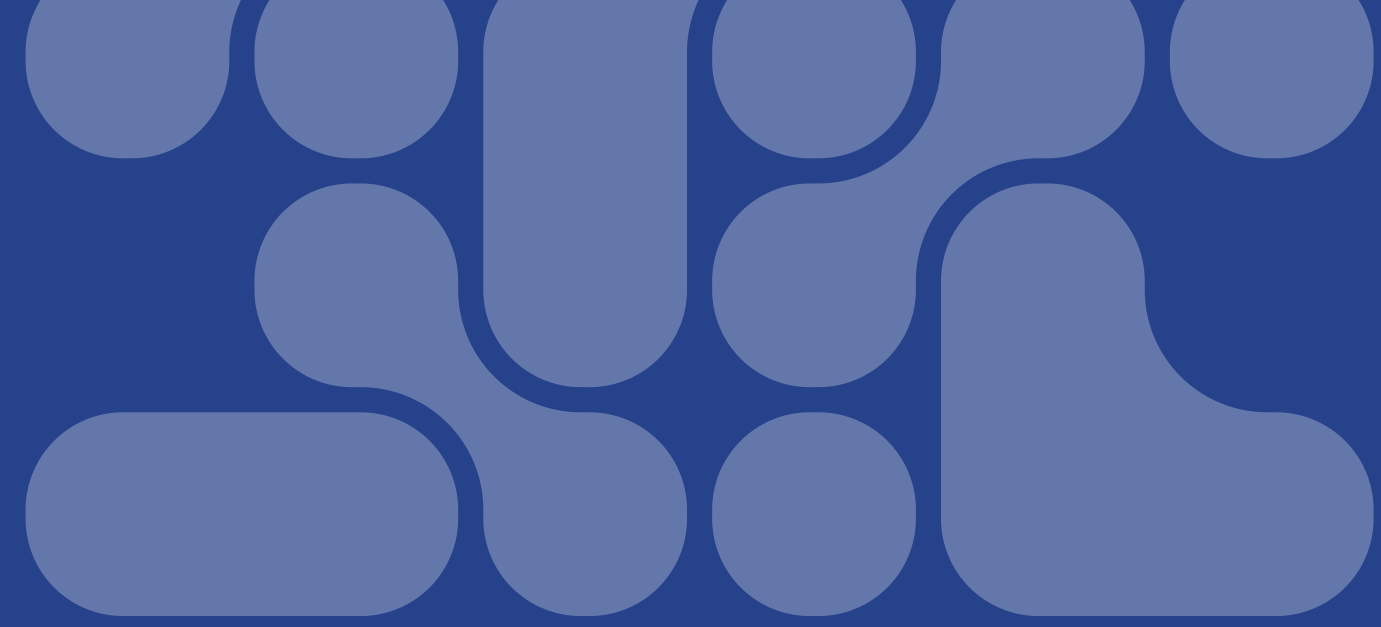
These methods use a model of the environment to predict outcomes and help the agent plan actions by simulating potential results.

- Markov decision processes (MDP)
- Bellman equation
- Value iteration algorithm
- Monte Carlo Tree Search

2. Model-Free Methods

The agent learns directly from experience by interacting with the environment and adjusting its actions based on feedback.

- Q-learning
- SARSA
- Monte Carlo Methods
- Reinforce algorithms
- Actor critic algorithm
- Asynchronous advantage actor-critic (A3C)



**Thank you for
your attention!**

Open floor for discussion.

